

DESIGN AND IMPLEMENTATION OF AN

AUTOMATED ANTENNA RADIATION PATTERN MEASUREMENT SYSTEM

Using Embedded Control and Image Processing

AOT

Academy of
Technology

MAKAUT, West Bengal

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Introduction — Problem Statement

WHY THIS PROJECT?

- Traditional antenna radiation pattern measurement relies on manual angular positioning — a process prone to human error and poor repeatability.
- Existing instruments require direct electrical connections to meters, introducing galvanic noise that distorts RF measurements.
- Most systems automate either rotation OR data acquisition — never both together in a unified embedded solution.
- Physical visualization of results still depends on manual plotting, introducing interpretation bias.
- Commercial RF measurement setups are prohibitively expensive for educational labs and small-scale research institutions.



[Fig. 1.1] — Existing Antenna Radiation Pattern Measurement Systems

OUR SOLUTION

A fully automated, low-cost, embedded system integrating stepper-motor rotation, optical OCR data acquisition, and CNC-based physical plotting for precise RF characterization.

OBJECTIVE

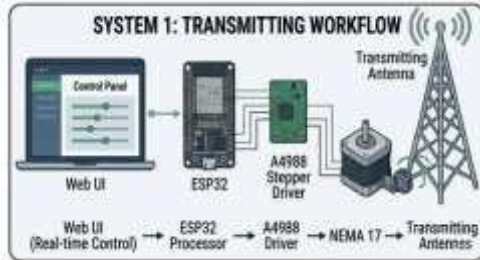
Automated
Rotation

OCR-based
Acquisition

CNC
Plotting

System Overview — Three Integrated Subsystems

System I: Antenna Rotation Control



[Fig. 2.1] — Transmitting System Workflow

- ESP32 drives NEMA 17 stepper motor via A4988 driver
- Precise 10° angular increments (36 positions/full sweep)
- 30-second field stabilization pause at each step
- Web UI-based real-time control (Start / Pause / Reset)

System II: Image Processing Unit



[Fig. 2.2] — Image Processing Unit Workflow

- ESP32S-CAM captures meter display images via OCR
- Creates an "optical bridge" — complete galvanic isolation
- Eliminates electrical noise and back-feeding distortion
- Extracted values transmitted wirelessly to ESP32 controller

System III: Physical Plotting System



[Fig. 2.3] — Plotting System Workflow

- 2-axis CNC plotter driven by 28BYJ-48 stepper motors
- ULN2003 driver boards control X and Y axes independently
- SG90 micro servo lifts/lowers pen for dot-free transitions
- Renders professional polar/Cartesian radiation graphs on A4 paper

Literature Survey — IEEE Referenced Works

[1]

R. C. Johnson (1993)

Antenna Engineering Handbook

Highlighted limitations of manual radiation pattern measurement; emphasized need for automated angular positioning to improve accuracy.

[2]

C. A. Balanis (2005)

Antenna Theory: Analysis and Design

Discussed radiation pattern characterization — noted traditional setups suffer from mechanical imprecision and human reading errors.

[3]

S. C. Orfanidis (2002)

Electromagnetic Waves and Antennas

Introduced digital methods in RF testing; measurement automation enhances repeatability in electromagnetic experiments.

[4]

J. Fraden (2010)

Handbook of Modern Sensors

Explored non-contact sensing — optical methods as reliable alternatives where direct electrical interface is not feasible.

[5]

Zhang et al. (2018)

Vision-Based Instrument Reading (IEEE Xplore)

Proposed OCR-based instrument reading to eliminate manual monitoring; image processing replaces wired data acquisition.
Link : <https://ieeexplore.ieee.org/document/8461184>

[6]

Y. Yalçın & S. Kurtulan (2009)

IEEE Antennas & Propagation Magazine

Motor-controlled antenna scanning improved angular precision but lacked automated data extraction capability.
Link : <https://doi.org/10.1109/MAP.2009.5162072>

Gap Analysis — Limitations in Existing Work

Identified Gap	Existing Limitation	Our Solution
No Fully Integrated Automation	Systems automate rotation OR measurement — never both.	Single ESP32 controller synchronizes rotation + data acquisition in one cycle.
Electrical Interface Dependency	Wired connections to meters introduce galvanic noise and back-feeding.	ESP32-CAM creates an 'optical bridge' — complete electrical isolation.
PC-Based Processing Requirement	Most setups rely on desktop computers for signal processing.	Fully embedded: ESP32 handles all calculations and coordination.
No Physical Graph Output	Graphs generated digitally but not drawn — require separate tools.	Onboard 2-axis CNC plotter produces hard-copy polar/Cartesian graphs.
Human Intervention Required	Operators must manually position, read meters, and plot results.	End-to-end automation from start command to finished radiation graph.

Theoretical Background

Antenna Radiation Fundamentals

Radiation patterns represent the directional energy distribution of an antenna, characterized by gain and beamwidth. Signal strength at each angular position is recorded and normalized to a reference value, then converted to dB: $P_{rel} = 20 \cdot \log_{10}(I/I_{max})$. [Ref: Balanis, 2005]

Embedded Automation & Motor Control

The ESP32 synchronizes a NEMA 17 stepper motor through an A4988 micro-stepping driver using GPIO 19 (STEP pulses) and GPIO 18 (DIR signals). 200 steps/rev motor with 1/16 micro-stepping can achieve sub-degree accuracy. SLP/RST pins are jumpered to maintain an active driver state.

Non-Contact OCR Data Acquisition

ESP32-CAM captures meter displays at high resolution. Edge-detection preprocessing isolates digit regions; Tesseract OCR extracts numerical values. This optical bridge provides complete galvanic isolation — eliminating back-EMF interference that would otherwise corrupt sensitive RF measurements. [Ref: Zhang et al., 2018]

CNC-Based Physical Plotting

Relative power values are mapped to rectangular plots. Two 28BYJ-48 steppers (ULN2003 driven) control X and Y axes. A GT2 timing belt and 300mm aluminium linear rails translate motor steps to precise pen displacements on A4 paper.

Hardware Requirements

Central Processing

- ESP32 (512MB RAM) — Main Controller & Web Server Host

Mechanical Rotation System

- NEMA 17 Stepper Motor — Rotates receiving antenna
- A4988 Motor Driver — Controls antenna rotation with micro-stepping

Computer Vision & Acquisition

- ESP32S-CAM — Captures high-res meter display images
- ESP32 CAM MB — Baseboard connector module (5V → 3.3V)

Physical Plotting System

- 28BYJ-48 Stepper Motor ×2 (X & Y axes)
- ULN2003 Driver Board ×2 (X-axis & Y-axis motor control)
- Linear Rails ×2 (300mm aluminium guides)
- GT2 Timing Belts with pulleys
- SG90 Micro Servo — Pen lift mechanism
- Spring-loaded Pen Holder & A4 Drawing Surface
- Acrylic/Aluminium Frame Structure

Power Supply

- 12V 4A Power Adapter
- USB-C connector for ESP32 (5V regulated to 3.3V)

Software Requirements & Development Stack

Arduino IDE

Firmware Development

Compiling and uploading embedded C++ firmware to ESP32 and ESP32-CAM modules via USB serial. Manages GPIO, WiFi, and HTTP server libraries.

Visual Studio Code

Application & Script Development

Python scripts for OCR post-processing.

Python 3.x

Data Processing Language

NumPy (numerical arrays), OpenCV (image preprocessing for OCR), Pytesseract (OCR engine wrapper). Runs on host machine for advanced processing.

Web Stack (HTML/CSS/JS)

Real-Time Web UI

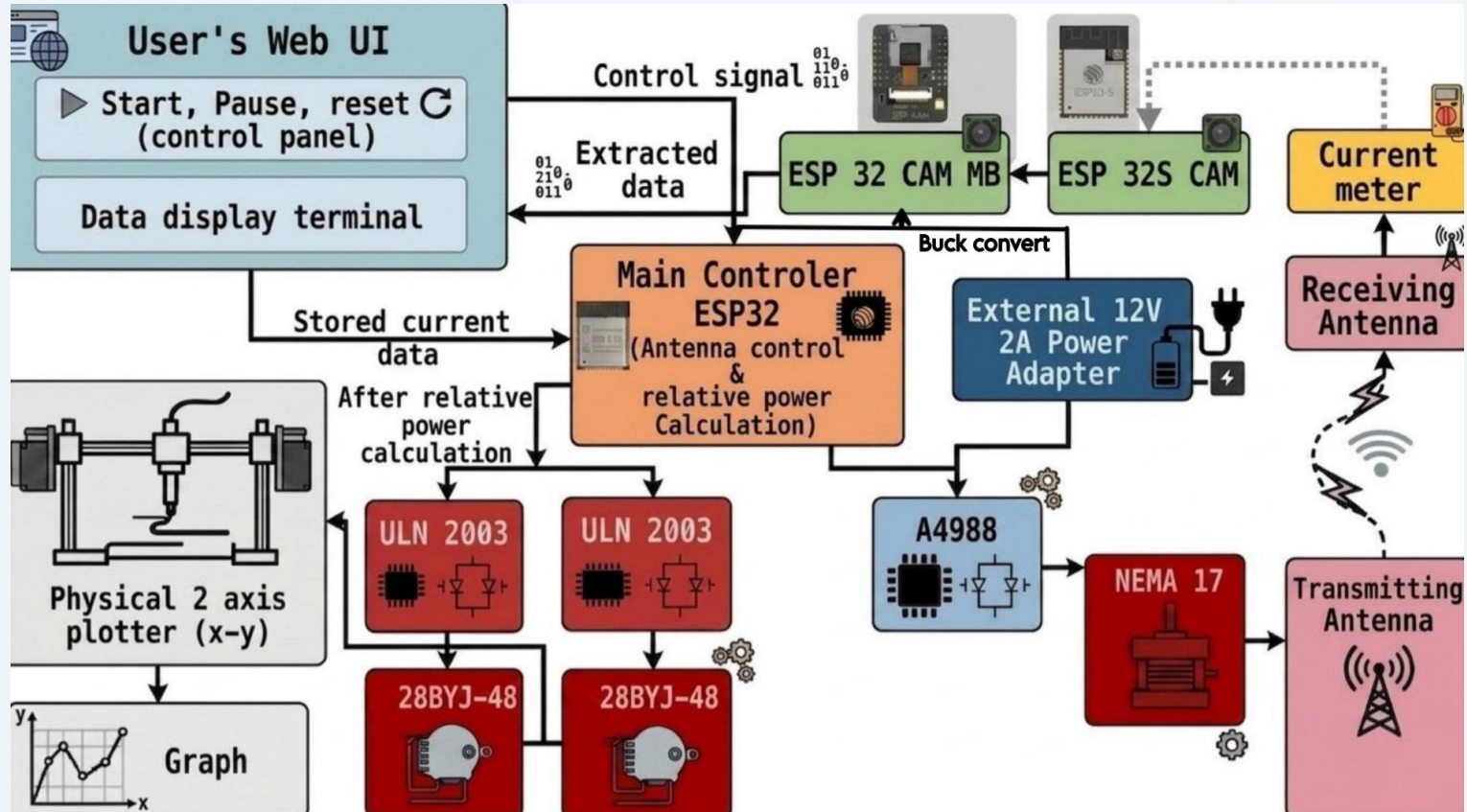
Mobile-friendly 'Antenna Command Center' served directly from ESP32 WebServer. Users control rotation, view live data, and monitor system status.

Libraries & Drivers

Low-Level Hardware Interface

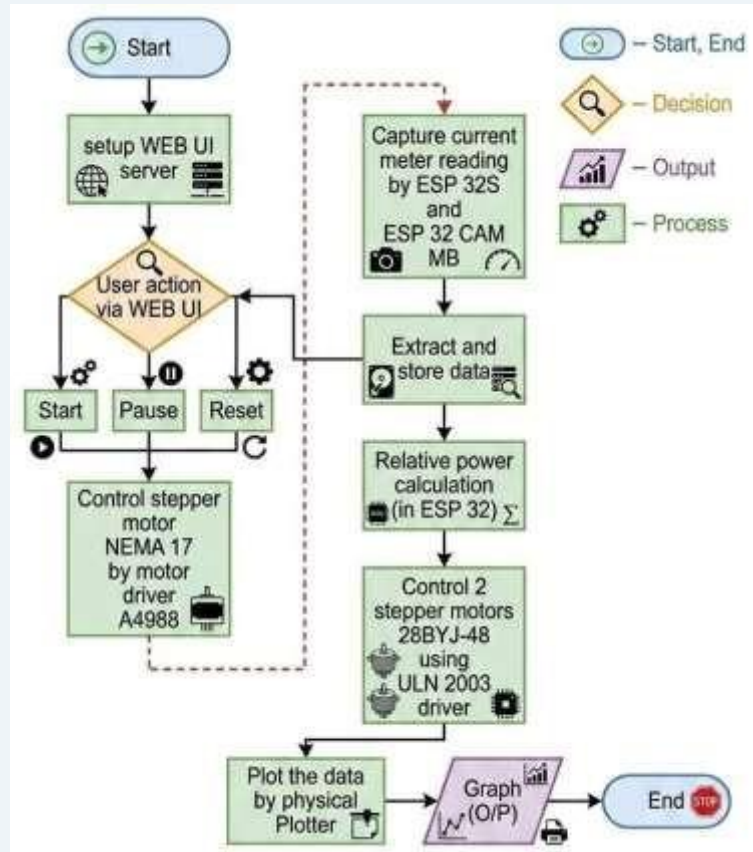
WiFi.h, esp_camera.h, esp_http_server — ESP32 core libraries. CP210X Windows Drivers for ESP 32. Pytesseract + Tesseract OCR engine for digit extraction.

System Block Diagram



[Fig. 6.1] — Complete system block diagram showing interconnects between Web UI, ESP32 controller, motor drivers, camera module, and CNC plotter.

System Workflow — End-to-End Operation



[Fig. 6.2] — Operational flowchart

1

Setup & Initialize

ESP32 boots, initializes Web UI server on local WiFi network.

2

User Command (Web UI)

Operator selects Start / Pause / Reset from mobile or browser interface.

3

Rotate Antenna (NEMA 17)

A4988 drives 10° step. System waits 30s for field stabilization.

4

Capture & OCR (ESP32-CAM)

ESP32S-CAM photographs meter; OCR extracts current reading (e.g., 14.7871 mA).

5

Calculate Relative Power

ESP32 computes: $P_{rel} = 20 \cdot \log_{10}(I/I_{max})$ in dB. Stores data point.

6

Repeat (36 positions)

Steps 3–5 repeat for each 10° increment across full 360° sweep.

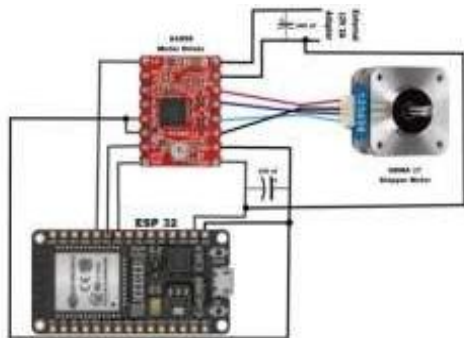
7

Physical CNC Plot

28BYJ-48 motors plot the radiation graph on A4 paper via CNC plotter.

Circuit Design — Antenna Rotation Control & Image Processing Unit

Rotation Control Circuit



[Fig. 7.1] — Rotation Control Circuit

Image Processing Setup



[Fig. 7.2] — Image Processing Setup

Circuit Specifications

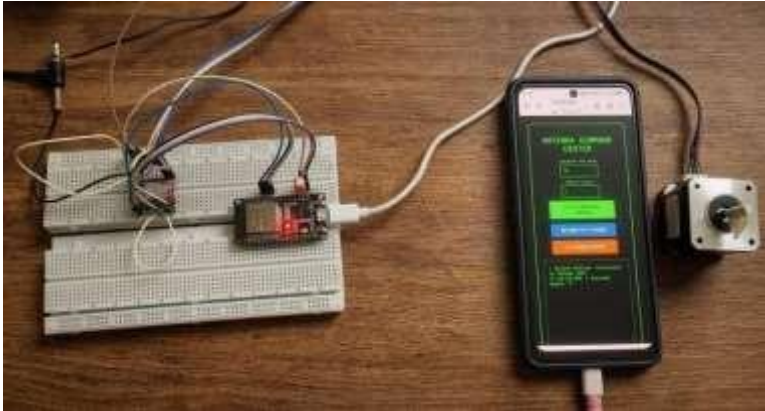
Power Supply:	External 12V 2A adapter → A4988 VMOT/GND
Capacitor:	100 μ F bulk capacitor for voltage stabilization
Logic Power:	3.3V + common ground from ESP32
GPIO 19:	STEP signal — controls stepper movement pulse
GPIO 18:	DIR signal — controls rotation direction
SLP / RST:	Jumpered HIGH to keep A4988 in active state
Coil 1:	Red / Deep Blue wires — NEMA 17 motor coil A
Coil 2:	Black / Sky Blue wires — NEMA 17 motor coil B

About Setup

- The USB cable provides a steady 5V supply from the laptop to the baseboard.
- The baseboard regulates this down to the 3.3V required by the ESP32 chip.

Prototype Development — Physical Implementation

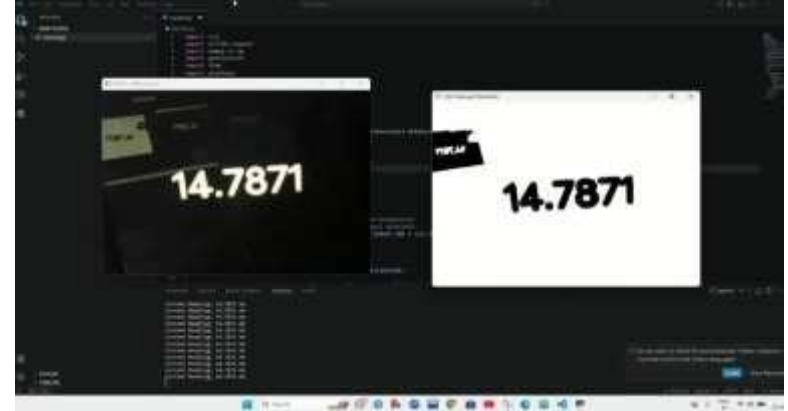
Part 1: Embedded Rotation Control & Web UI



[Fig. 8.1] — ESP32 breadboard prototype with NEMA 17 stepper and mobile Web UI

- ESP32 hosts web server on local WiFi ('Antenna Command Center')
- A4988 drives NEMA 17 in precise 10° increments
- 30-second stabilization pause ensures consistent field readings
- Real-time status visible on mobile browser

Part 2: Computer Vision & Data Acquisition



[Fig. 8.2] — ESP32-CAM capturing '14.7871 mA' from meter display via OCR

- High-resolution images captured of digital meter readout
- OCR extracts precise values: e.g., '14.7871 mA' demonstrated
- Complete galvanic isolation — no back-feeding noise
- Data transmitted wirelessly back to main ESP32 controller

Milestones Achieved — System Analysis



Antenna Rotation Control

NEMA 17 stepper motor rotation implemented with 10° precision. Successfully demonstrates full 360° sweep without manual intervention.

COMPLETE



Web-Based Control Interface

Functional mobile-friendly web UI developed and tested. Real-time motor control, start/pause/reset operations verified on local WiFi server.

COMPLETE



Non-Contact Data Acquisition

ESP32-CAM + OCR pipeline integrated. Numerical values extracted from meter display autonomously — '14.7871 mA' successfully read in live demo.

COMPLETE



Physical CNC Plotter (In Progress)

28BYJ-48 motor control and ULN2003 driver integration underway. Polar graph rendering algorithm developed; hardware assembly in final stage.

IN PROGRESS

Future Scope & Target Market

Future Enhancements

Polar Graph Plotting

Transition from Cartesian to true polar coordinate hard-copy output to better represent standard radiation pattern conventions.

AI-Enhanced OCR

Integrate a trained deep-learning model to improve OCR reliability across different meter display types and lighting conditions.

Multi-Frequency Support

Extend measurement capability across multiple RF bands by integrating a software-defined radio (SDR) frontend.

3D Radiation Pattern

Add an elevation-plane scanning axis for full 3D spherical radiation pattern mapping and visualization.

Target Market & Customers

01 IoT & Wireless Developers

Engineers characterizing antenna performance for IoT sensors, WiFi modules, and LPWAN devices who need precise directional data without expensive professional equipment.

02 Automated RF Research Labs

Research facilities seeking to eliminate manual observation errors and integrate continuous, unattended antenna measurement into electromagnetic test workflows.

03 Educational Institutions

Universities and technical colleges requiring affordable, hands-on RF measurement training kits that demonstrate antenna theory concepts physically. Ideal for MAKAUT/ECE labs.

CONCLUSION

Project Summary & Key Achievements

✓ Antenna Rotation Control

Precise 10° automated stepper rotation achieved using ESP32 + NEMA 17 + A4988 — fully replacing manual angular positioning.

✓ Web-Based Real-Time UI

Mobile 'Antenna Command Center' web interface developed on ESP32 WebServer — operational for live motor control and monitoring.

✓ Non-Contact OCR Acquisition

ESP32-CAM+ Tesseract OCR pipeline successfully extracts meter readings (e.g., 14.7871 mA) with complete galvanic isolation.

The complete module will demonstrate the feasibility, accuracy, and practical applicability of the proposed embedded automation system.

References — IEEE Format

[1]

R. C. Johnson, Antenna Engineering Handbook, 3rd ed. New York, NY: McGraw-Hill, 1993.

[2]

C. A. Balanis, Antenna Theory: Analysis and Design, 3rd ed. Hoboken, NJ: Wiley-Interscience, 2005.

[3]

S. C. Orfanidis, Electromagnetic Waves and Antennas. New Brunswick, NJ: Rutgers University, 2002.

[4]

J. Fraden, Handbook of Modern Sensors: Physics, Designs, and Applications, 4th ed. New York, NY: Springer, 2010.

[5]

Zhang et al., 'Vision-based instrument reading using image processing and OCR techniques,' in Proc. IEEE Int. Conf. Systems, Man, Cybernetics, 2018.
Link : <https://ieeexplore.ieee.org/document/8461184>

[6]

Y. Yalçın and S. Kurtulan, 'Motor-controlled antenna scanning system for radiation pattern measurement,' IEEE Antennas Propag. Mag., vol. 51, no. 3, pp. 147–155, Jun. 2009.
Link : <https://doi.org/10.1109/MAP.2009.5162072>

THANK YOU

Questions & Discussion

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